# ORF307
# The implementation of the Simplex method

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## *Today's topics*

- Finding an initial basic feasible solution
- Degeneracy
- Full Simplex example
- Efficiency

- **Recap**

# *Standard form polyhedra*

Consider the LP in standard form:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

**Assumptions**:

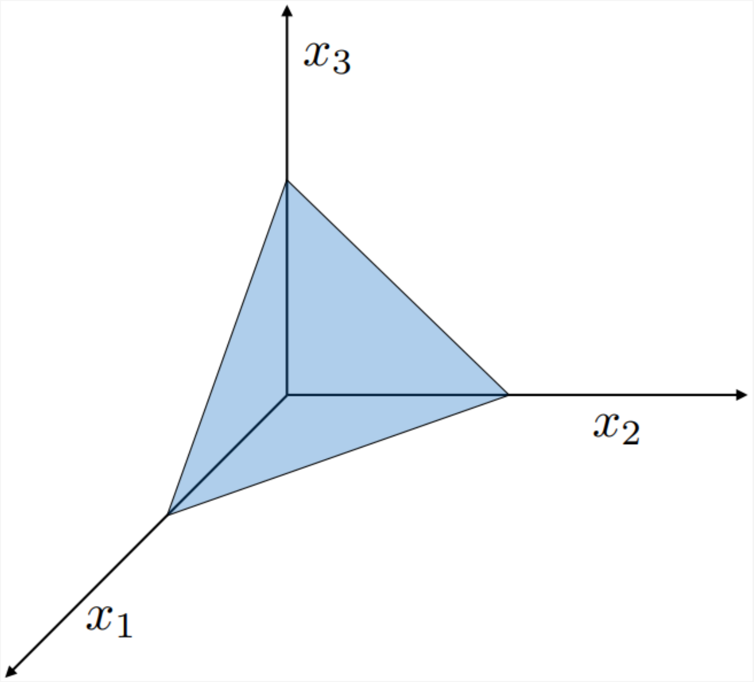
$A \in R^{m \times n}$ has full **row** rank and $m \leq n$

**Interpretation**:

$P$ is an $(n-m)$-dimensional surface

The feasible region is a standard polyhedron

$$P = \{x \in R^n \mid Ax = b, x \geq 0\}$$




Example with $m = 1$ and $n = 3$

# *An iteration of the Simplex method*

**Initialization:**

- A basic feasible solution $x$
- A basis matrix $A_B = \begin{bmatrix} A_{B(1)} & \dots & A_{B(m)} \end{bmatrix}$

**Iterations:**

1. Compute the reduced costs $\bar{c}$
   - Solve $A_B^T p = c_B$
   - $\bar{c} = c - A^T p$
2. If $\bar{c} = 0$, STOP. Declare $x$ optimal.
3. Choose $j$ such that $c_j < 0$
4. Compute search direction $d$ with

   $d_j = 1$ and $A_B d_B = -A_j$
5. If $d_B \geq 0$, the problem is unbounded. STOP.
6. Compute stepsize: $\theta^* = \min_{\{i \in B \mid d_i < 0\}} \left( -\frac{x_i}{d_i} \right)$
7. Define new vertex $y = x + \theta^* d$
8. Get new Basis $\bar{B}$ ($i$ exits and $j$ enters)

# *Example*

$$P = \{x \in R^3 \mid x_1 + x_2 + x_3 = 2, x \geq 0\}$$

$$x = (2,0,0)^T \qquad \boldsymbol{B} = \{\mathbf{1}\}$$

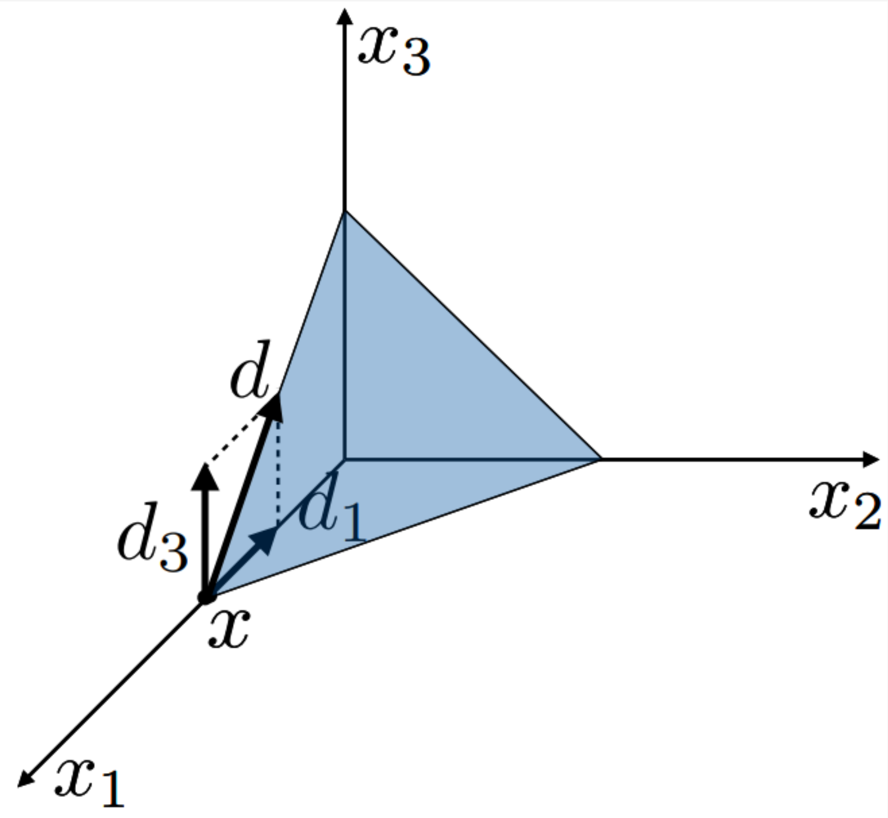
**Basic index**: $j = 3$ → $d = (-1,0,1)^T, \quad d_j = 1$

$$A_B d_B = -A_j \quad \Rightarrow \quad d_B = -1$$

**Stepsize**:

$$\theta^* = -\frac{x_1}{d_1} = -\frac{2}{-1} = 2$$

**New solution**: $y = x + \theta^* d = (0,0,2)^T \qquad \overline{\boldsymbol{B}} = \{\mathbf{3}\}$



# *Finite convergence theorem*

**IF**

- $P = \{x \mid Ax = b, x \geq 0\} \neq \emptyset$
- Every basic solution is **non-degenerate**

**THEN**

- The Simplex algorithm **terminates after a finite number of iterations**
- At termination we **either** have one of the following results
  - An **optimal** basis $B$ (i.e., the algorithm found the optimal solution)
  - A direction $d$ such that $Ad = 0,\ d \geq 0,\ c^T d < 0$ and optimal cost is $-\infty$ (**unbounded**)

# *Finite convergence theorem*

**Proof sketch:**

**The fact is:** at each iteration, the Simplex algorithm **improves (reduces) the objective** →

→ by a positive amount $\theta^*$, along the direction $d$ such that $c^T d < 0$

Therefore, the following two statements hold:

- The cost strictly decreases
- No basic feasible solution can be visited twice

Since there is only a **finite number of basic feasible solutions (vertices)**, the Simplex algorithm must eventually terminate. ■

# Find an initial point

# *Initial basic feasible solution*

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

**Question**: How can we compute an initial basic feasible solution $x$ and a basis $B$ ?

**Question**: Does it exist?

# *Initial basic feasible solution*

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned} \quad \Rightarrow \quad \begin{aligned} \min \quad & \mathbf{1}^T y \\ \text{s.t.} \quad & Ax + y = b \\ & x \geq 0, y \geq 0 \end{aligned}$$

Violations of the
Equality constraints

- W.L.O.G we can assume: $b \geq 0$ (if not true, multiply by $-1$)
- Trivial (and easy to compute) **feasible solution**: $x = 0, y = b$

**Possible outcomes**:

- **Feasible LP (cost = 0):** $y^* = 0$ and $x^*$ is basic feasible solution
- **Infeasible LP (cost > 0):** $y^* > 0$ gives us the violations of the equality constraints

# *Two-phase Simplex method*

**Phase I:**

1. Construct the auxiliary problem, such that $b \geq 0$
2. Solve the auxiliary problem using Simplex method starting from $(x, y) = (0, b)$
3. If the optimal value is greater than $0$ ($\mathbf{1}^T y > 0$), the original LP is infeasible → STOP

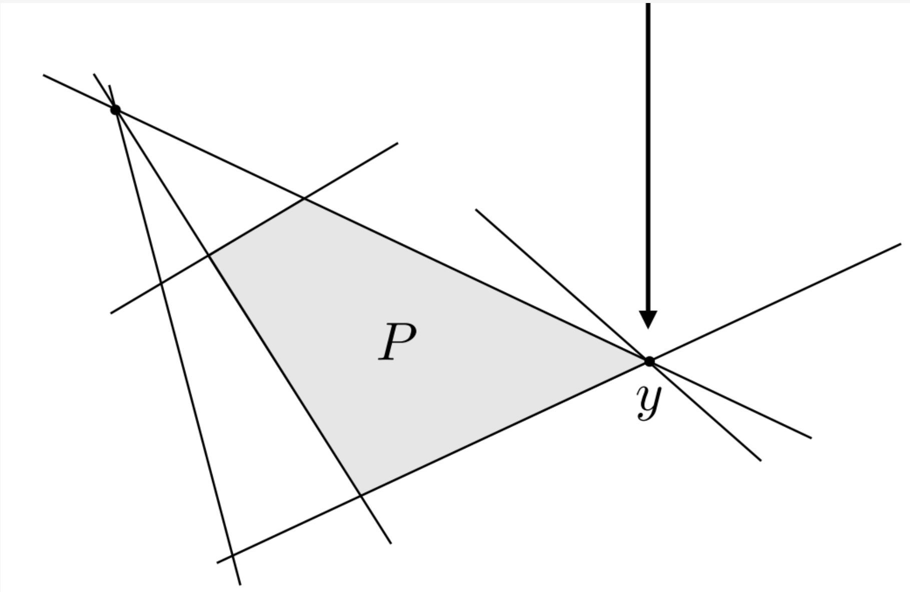
**Phase II:**

1. Recover original LP (i.e., drop $y$ and restore original cost)
2. Solve original LP starting from the optimal solution of the auxiliary LP and its basis

# Degeneracy

# *Degenerate basic feasible solutions*

A solution $y$ is degenerate if $|I(y)| > n$




**Inequality form polyhedron:**

$P = \{x \mid Ax \leq b\}.$

# *Degenerate basic feasible solutions*

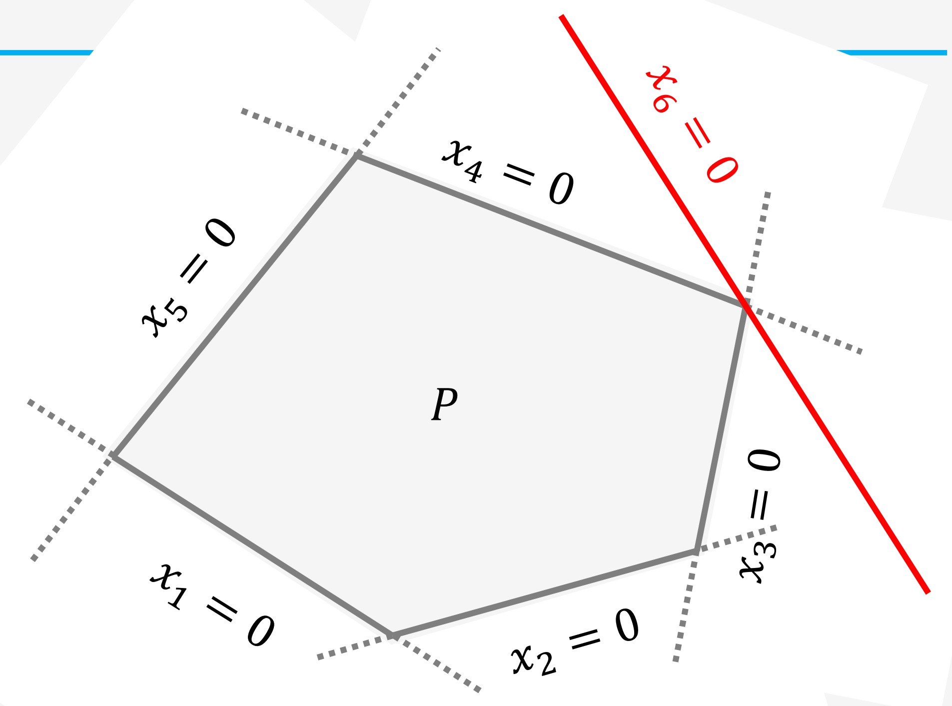
**Standard form polyhedron:**

$$P = \{x \mid Ax = b, x \geq 0\}.$$

Given a basis matrix $A_B = [A_{B(1)}, A_{B(2)}, \ldots, A_{B(m)}]$,

We can compute a **Basic Feasible Solution**

- $A_B x_B = b$
- $x_i = 0, \ \forall i \neq B(1), B(2), \ldots, B(m)$




**Remark**: if **some elements** of $x_B$ are **zero**, then it is a **degenerate solution**.

# *Degenerate basic feasible solutions*

**Example:**

Consider the feasible region defined by:

$$\begin{aligned} x_1 + x_2 + x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 1 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

**Degenerate solutions**

$$\begin{aligned} B = \{\mathbf{1}, \mathbf{2}\} \quad &\rightarrow \quad x = (\mathbf{0}, \mathbf{1}, \quad 0)^T \\ B = \{\mathbf{2}, \mathbf{3}\} \quad &\rightarrow \quad x = (0, \quad \mathbf{1}, \mathbf{0})^T \end{aligned}$$

# *Cycling*

**Step-size computation:** recall **Step 6** of Simplex algorithm:

**6**. compute step length

$$\theta^* = \min_{\{i \in B \mid d_i < 0\}} \left( -\frac{x_i}{d_i} \right)$$

If $i \in B$ and $d_i < 0$ and $x_i = 0$, then we will have $\theta^* = 0$. (degenerate case)

**Hence**: $y = x + \theta^* x = x$ → same solution → **we are not jumping to a neighboring vertex**

$B \neq \bar{B}$ → different basis

**Finite termination is no longer guaranteed!** → How can we fix that?

Pivoting rule →...

# *Pivoting rules*

**Choose the index that will enter the basis:**

Simplex iteration:

- **Step 3**: choose $j$ such that $\bar{c}_j < 0$ → but which $j$ is the best to select?

## Possible selection rules

- **Smallest subscript**: smallest $j$ such that $\bar{\boldsymbol{c}}_{\boldsymbol{j}} < \mathbf{0}$
- **Most negative**: choose $j$ with the most negative $\bar{c}_j$
- **Largest cost decrement**: choose $j$ with the largest $\theta^* |\bar{\boldsymbol{c}}_{\boldsymbol{j}}|$

# *Pivoting rules*

**Choose the index that will exit the basis:**

Simplex iteration:

- Step 6: compute the step length

$$\theta^* = \min_{\{i \in B \mid d_i < 0\}} \left( -\frac{x_i}{d_i} \right)$$

Index $i$ that corresponds to the smallest ratio will exit the basis

**NOTE**: We can have more than one $i$ for which $x_i = 0$ → So, the next solution is degenerate!

**For example:** we may have $x_4 = 0$ and $x_{17} = 0$

Which $i$ should we select in this case? → Smallest $i$ such that $\theta^* = -\frac{x_i}{d_i} = 0$ → $i = 4$

$x_4$ exits the Basis

# *Bland's rule to avoid cycling*

**Theorem:**

If we use the **smallest index rule** for choosing both the index $j$ entering the basis, and the index $i$ exiting the basis, **then no cycling will occur**.

**Proof**: see [Vanderbei, Ch. 3, Sec. 4], [Bertsimas/Tsitsiklis, Sec. 3.4]

Main idea:

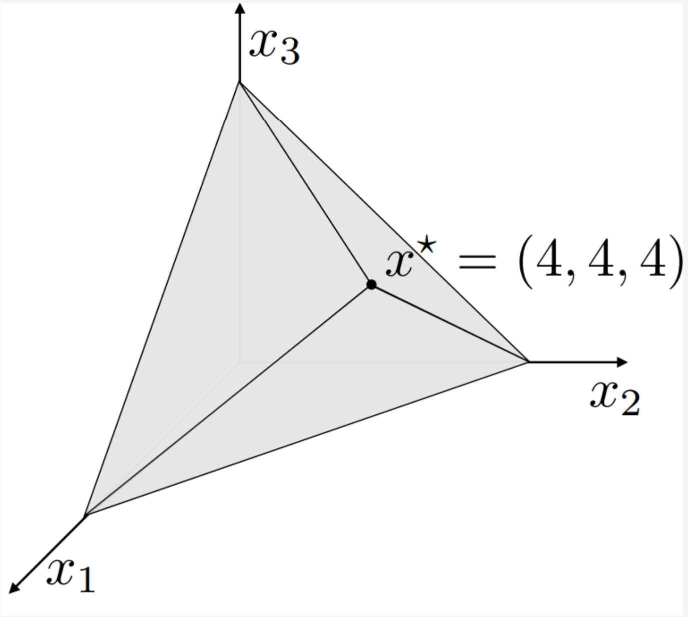
Assume Bland's rule is applied and there is no cycling with different bases

Obtain a contradiction

# Simplex iterations – Example

# *Example*

**LP in inequality form**: $$\begin{aligned} \min\; & -10x_1 - 12x_2 - 12x_3 \\ \text{s.t.}\quad & x_1 + 2x_2 + 2x_3 \leq 20 \\ & 2x_2 + x_2 + x_3 \leq 20 \\ & 2x_1 + 2x_2 + x_3 \leq 20 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$




**LP in standard form**: $\min -10x_1 - 12x_2 - 12x_3$

(adding slack variables)

$$\text{s.t.} \quad \begin{bmatrix} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix} = \begin{pmatrix} 20 \\ 20 \\ 20 \end{pmatrix}$$

$$x \geq 0$$

# *Example – initialization*

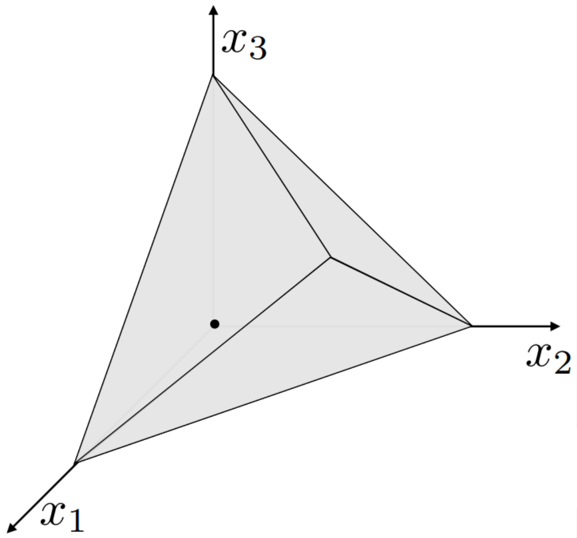
$$c = (-10, -12, -12,0,0,0)^T$$

LP in inequality form:

$$\begin{aligned} \min\ & -10x_1 - 12x_2 - 12x_3 \\ \text{s.t.}\ & x_1 + 2x_2 + 2x_3 \leq 20 \\ & 2x_2 + x_2 + x_3 \leq 20 \\ & 2x_1 + 2x_2 + x_3 \leq 20 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$b = (20,20,20)^T$$




**Initial point and Basis:** $x = (0,0,0,20,20,20)^T$

Basis: $B = \{4,5,6\}$

$$A_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

# Example – iteration #1

$c = (-10, -12, -12,0,0,0)^T$

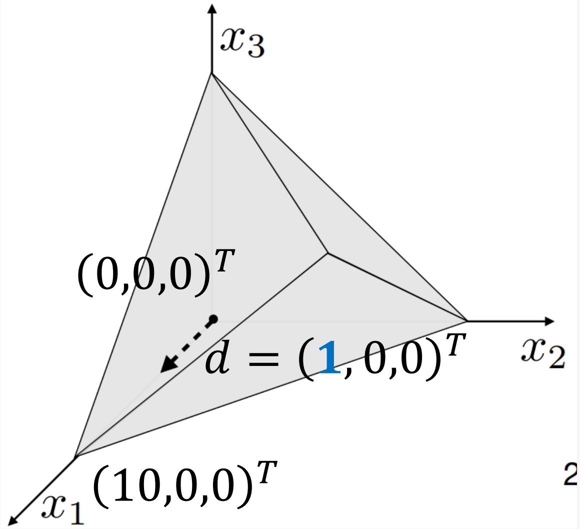
$$\begin{aligned}\min\; & -10x_1 - 12x_2 - 12x_3 \\ \text{s.t.}\; & x_1 + 2x_2 + 2x_3 \le 20 \\ & 2x_2 + x_2 + x_3 \le 20 \\ & 2x_1 + 2x_2 + x_3 \le 20 \\ & x_1, x_2, x_3 \ge 0\end{aligned}$$

**Current point**:
$x = (0,0,0,20,20,20)^T$
$c^T x = 0$
Basis: $B = \{4,5,6\}$

$$A_B = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$A = \begin{bmatrix}1 & 2 & 2 & 1 & 0 & 0\\ 2 & 1 & 2 & 0 & 1 & 0\\ 2 & 2 & 1 & 0 & 0 & 1\end{bmatrix}$$

$b = (20,20,20)^T$




**Reduced costs**: $\bar{c} = ?\,,\;\; c_B = 0$
Solve $A_B^T p = c_B \;\Rightarrow\; p = c_B = 0$
$\bar{c} = c - A^T p = c = (-10, -12, -12) \le 0.$

**Direction**: Solve: $A_B d_B = -A_j \;\Rightarrow\; d_B = (-1, -2, -2)^T$
$j = 1$ **enters** the basis,
Set: $d = (1, 0,0, -1, -2, -2)^T$

**Stepsize**: $\theta^* = 10,\;\; i = 5$
$\theta^* = \min_{\{i \in B | d_i < 0\}} (-x_i/d_i) = \min\{20, \mathbf{10}, 10\} = 10 \Rightarrow i = 5$ **exits**
New point: $x \leftarrow x + \theta^* d = (10,0,0,10,0,0)^T$

**New basis**: $B = \{4, 1, 6\}$

$$A_B = \begin{bmatrix}1 & 1 & 0\\ 0 & 2 & 0\\ 0 & 2 & 1\end{bmatrix}$$

# Example – iteration #2

$c = (-10, -12, -12,0,0,0)^T$

$$\begin{aligned} \min\; & -10x_1 - 12x_2 - 12x_3 \\ \text{s.t.}\; & x_1 + 2x_2 + 2x_3 \le 20 \\ & 2x_2 + x_2 + x_3 \le 20 \\ & 2x_1 + 2x_2 + x_3 \le 20 \\ & x_1, x_2, x_3 \ge 0 \end{aligned}$$

**Current point**:

$x = (10,0,0,10,0,0)^T$

$c^T x = -100$

Basis: $B = \{4, \mathbf{1}, 6\}$

$$A_B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

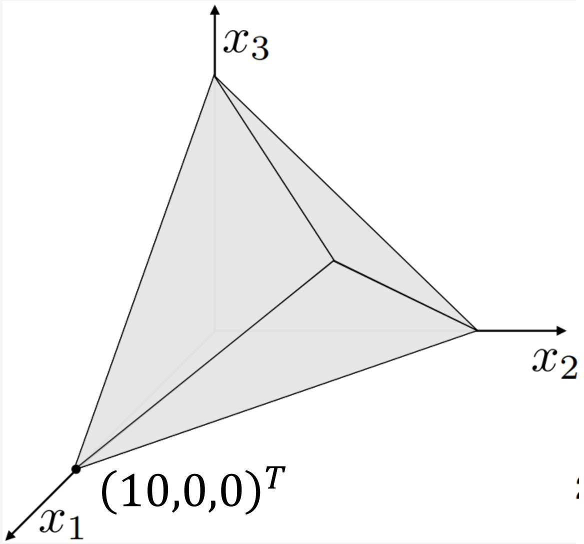
$$A = \begin{bmatrix} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$b = (20,20,20)^T$

**Reduced costs**: $\bar{c} = ?\,,\;\; c_B = (0, -10, 0)$

Solve $A_B^T p = c_B \;\Rightarrow\; p = (0, -5, 0)^T$

$\bar{c} = c - A^T p = (0, -7, -2, 0, 5, 0)^T$.




**Direction**: Solve: $A_B d_B = -A_j \;\Rightarrow\; d_B = (-1.5, -0.5, -1)^T$

$\boldsymbol{j = 2}$ **enters** the basis,

Set: $d = (-0.5, 1, 0, -1.5, 0, -1)^T$

**Stepsize**: $\theta^* = 0$, $i = 6$

$$\theta^* = \min_{\{i \in B \mid d_i < 0\}} (-x_i / d_i) = \min\{6.66, 20, \mathbf{0}\} = \mathbf{0} \Rightarrow i = 6 \textbf{ exits}$$

New point: $x \leftarrow x + \theta^* d = (10,0,0,10,0,0)^T$ ⟵ **Cycling**

New: $B = \{4, 1, 2\}$

$$A_B = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

# Example – *iteration #3*

$c = (-10, -12, -12, 0, 0, 0)^T$

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$b = (20, 20, 20)^T$

$$\begin{aligned} \min\ & -10x_1 - 12x_2 - 12x_3 \\ \text{s.t.}\ & x_1 + 2x_2 + 2x_3 \le 20 \\ & 2x_2 + x_2 + x_3 \le 20 \\ & 2x_1 + 2x_2 + x_3 \le 20 \\ & x_1, x_2, x_3 \ge 0 \end{aligned}$$

**Current point**:

$x = (10, 0, 0, 10, 0, )^T$

$c^T x = -100$

Basis: $B = \{4, 1, 2\}$

$$A_B = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

**Reduced costs**: $\bar{c} = ?\quad c_B = (0, -10, -12)$

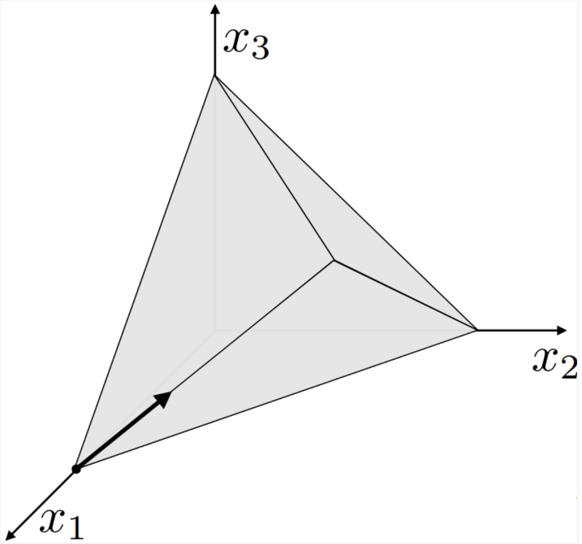
Solve $A_B^T p = c_B \ \Rightarrow\ p = (0, 2, -7)^T$

$\bar{c} = c - A^T p = (0, 0, -9, 0, -2, 7)^T$.

**Direction**: Solve: $A_B d_B = -A_j \ \Rightarrow\ d_B = (-2.5, -1.5, 1)^T$

$j = 3$ **enters** the basis,

Set: $d = (-1.5, 1, 1, -2.5, 0, 0)^T$




**Stepsize**: $\theta^* = 4,\ \ i = 4$

$\theta^* = \min_{\{i | d_i < 0\}} (-x_i / d_i) = \min\{4, 6.67\} = 4 \Rightarrow i = 4$ **exits**

New point: $x = x + \theta^* d = (4, 4, 4, 0, 0, 0)^T$

New: $B = \{3, 1, 2\}$

$$A_B = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

# *Example – iteration #4*

$c = (-10, -12, -12, 0,0,0)^T$

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$b = (20,20,20)^T$

$$\begin{aligned} \min\ & -10x_1 - 12x_2 - 12x_3 \\ \text{s.t.}\ & x_1 + 2x_2 + 2x_3 \le 20 \\ & 2x_2 + x_2 + x_3 \le 20 \\ & 2x_1 + 2x_2 + x_3 \le 20 \\ & x_1, x_2, x_3 \ge 0 \end{aligned}$$

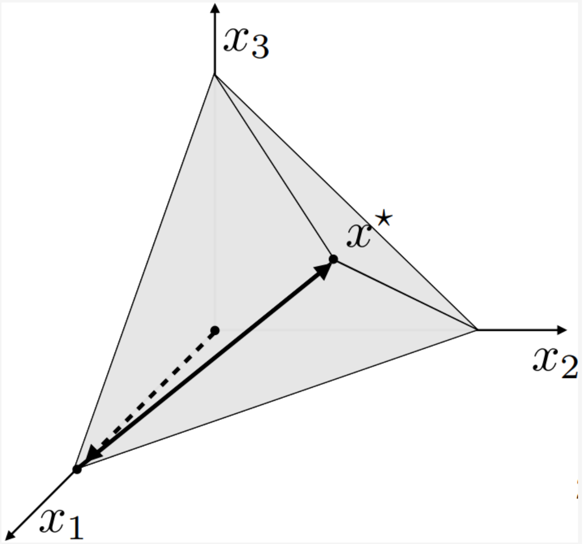
**Current point**:

$x = (4,4,4,0,0,0,)^T$

$c^T x = -136$

Basis: $B = \{3,1,2\}$

$$A_B = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$




**Reduced costs**: $\bar{c} =?$ $\quad c_B = (-12, -10, -12)$

Solve $A_B^T p = c_B \ \Rightarrow \ p = (-3.6, -1.6, -1.6)^T$

$\bar{c} = c - A^T p = (0,0,0,3.6,1.6,1.6)^T$.

$\bar{c}_1, \bar{c}_2, \bar{c}_3 \ge 0$

**OPTIMAL**: $\bar{c} \ge 0 \quad \to \quad x^* = (4,4,4,0,0,0)^T$

# Complexity

# *Complexity of a single Simplex iteration*

**Iteration:**

1. Compute the reduced costs $\bar{c}$
   - Solve $A_B^T p = c_B$
   - $\bar{c} = c - A^T p$
2. If $\bar{c} = 0$, STOP. Declare $x$ optimal.
3. Choose $j$ such that $c_j < 0$
4. Compute search direction $d$ with
   $d_j = 1$ and $A_B d_B = -A_j$
5. If $d_B \geq 0$, the problem is unbounded. STOP.
6. Compute stepsize
7. Define $y$ such that $y = x + \theta^* d$
8. Get new Basis $\bar{B}$ ($i$ exits and $j$ enters)

**Bottlenecks**

**Solution of two linear systems**

# *Solution of the two linear systems*

**Very similar systems**

$$A_B^T p = c_B$$

$$A_B d_B = -A_j$$

**$LU$ factorization**

**$(2/3)n^3$ flops**

$$A_B = PLU$$

**Easy linear systems**

**$4n^2$ flops**

$$U^T L^T P^T p = c_B$$

$$PLU d_B = -A_j$$

**Factorization is computational expensive!**

**Do we need to recompute at every Simplex iteration?**

# *Basis update*

**Index update:**

- $j$ enters ($x_j$ becomes $\theta^*$)
- $i = B(\ell)$ exits ($x_i$ becomes 0)

**Basis matrix update:**

$$A_{\bar{B}} = A_B + \left(A_{\boldsymbol{i}} - A_{\boldsymbol{j}}\right)e_\ell^T$$

**Example**

$$A = \begin{array}{c} \begin{bmatrix} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{bmatrix} \\ \begin{matrix} \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} & \mathbf{5} & \mathbf{6} \end{matrix} \end{array}$$

$B = \{4,1,6\} \quad \rightarrow \quad \bar{B} = \{4,1,2\}$

- 2 enters the basis
- $6 = B(3)$ exists the basis

$$A_{\bar{B}} = \begin{array}{c} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 1 \\ 0 & 2 & 2 \end{bmatrix} \\ \begin{matrix} \mathbf{4} & \mathbf{1} & \mathbf{2} \end{matrix} \end{array} = \underbrace{\begin{array}{c} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 1 \end{bmatrix} \\ \begin{matrix} \mathbf{4} & \mathbf{1} & \mathbf{6} \end{matrix} \end{array}}_{A_B} + \underbrace{\begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix}}_{A_{\mathbf{2}} e_3^T} - \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{A_{\mathbf{6}} e_3^T}$$

# *Smarter linear system solution*

**Basic matrix change:**

$$A_{\bar{B}} = A_B + \overbrace{(A_i - A_j)}^{v} e_\ell^T \quad \Longrightarrow$$

**Matrix inversion lemma**

$$\left(A_B + v e_\ell^T\right)^{-1} = \left(I - \frac{1}{1 + e_\ell^T A_B^{-1} v} A_B^{-1} v e_\ell^T\right) A_B^{-1}$$

**You proved this in HW2**

**Solve:** $A_{\bar{B}} d_{\bar{B}} = -A_j$

1. Solve: $A_B z^1 = e_\ell$ ($2n^2$ flops)
2. Solve: $A_B z^2 = -A_j$ ($2n^2$ flops)
3. Solve: $d_{\bar{B}} = z^2 - \frac{v^T z^2}{1 + v^T z^1} z^1$

**Remarks:**

- Same complexity for $A_B^T p = c_B$ ($4n^2$ flops)
- $k$-th next iteration ($4kn^2$ flops → Exercise!)
- After certain number of iteration, refactor $A_B$ (e.g., after $k = 100$ iterations)

# *Complexity of a single Simplex iteration*

**Iteration:**

1. Compute the reduced costs $\bar{c}$
   - Solve $A_B^T p = c_B$
   - $\bar{c} = c - A^T p$
2. If $\bar{c} = 0$, STOP. Declare $x$ optimal.
3. Choose $j$ such that $c_j < 0$
4. Compute search direction $d$ with
   $d_j = 1$ and $A_B d_B = -A_j$
5. If $d_B \geq 0$, the problem is unbounded. STOP.
6. Compute stepsize
7. Define $y$ such that $y = x + \theta^* d$
8. Get new Basis $\bar{B}$ ($i$ exits and $j$ enters)

**Bottlenecks**

**Solution of two linear systems**

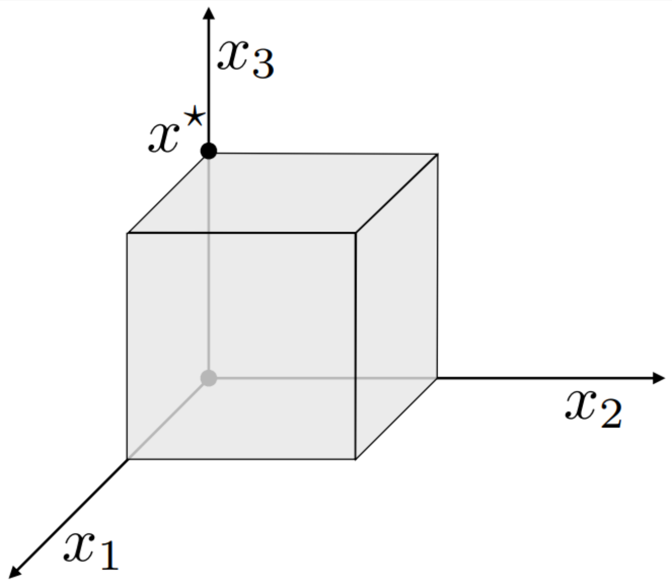
**Matrix inversion trick**

$\approx n^2$ flops per iterations

(very inexpensive)

# *Complexity of the Simplex method*

**Worst case behavior of Simplex:**




Consider the very simple LP:

$$\begin{aligned} \min \quad & -x_n \\ \text{s.t.} \quad & 0 \le x_i \le 1, \;\; i = 1, \dots, n \end{aligned}$$

**Characteristics**:

$2^n$ vertices

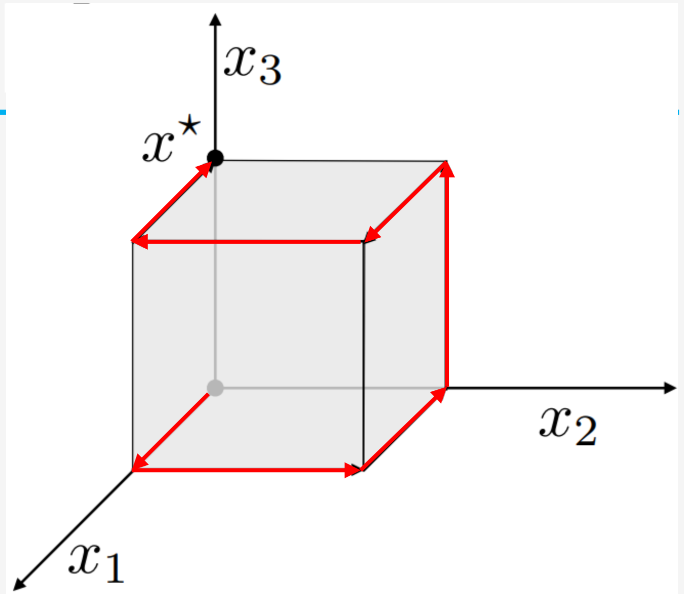
$2^n/2$ vertices have cost=0

$2^n/2$ vertices have cost=1

**Perturbed unit hypercube:**

$$\begin{aligned} \min \quad & -x_n \\ \text{s.t.} \quad & \epsilon \le x_1 \le 1 \\ & \epsilon x_{i-1} \le x_i \le 1 - \epsilon x_{i-1}, \;\; i = 2, \dots, n \end{aligned}$$

# *Complexity of the Simplex method*




**Perturbed unit hypercube:**

$$
\begin{aligned}
\min \quad & -x_n \\
\text{s.t.} \quad & \epsilon \leq x_1 \leq 1 \\
& \epsilon x_{i-1} \leq x_i \leq 1 - \epsilon x_{i-1}, \;\; i = 2, \dots, n
\end{aligned}
$$

**Theorem:**

- the vertices can be ordered so that each one is adjacent to and has a lower cost than the previous
- There exists a pivoting rule under which the Simplex method terminates after $2^{n-1}$ iterations

**Remark**:

- A different pivoting rule would have converged in 1 iteration
- We have a bad example for every pivoting rule

# *Complexity of the Simplex method*

**Is the Simplex method efficient algorithm?**

We do not any polynomial version of the Simplex method, no matter which pivoting rule we pick.

Showing the Simplex is Polynomial time algorithm is an open research problem.

**Worst-case:**

There are problem instances where Simplex performs exponentially many iterations (e.g., $2^n$)

**Good news:**

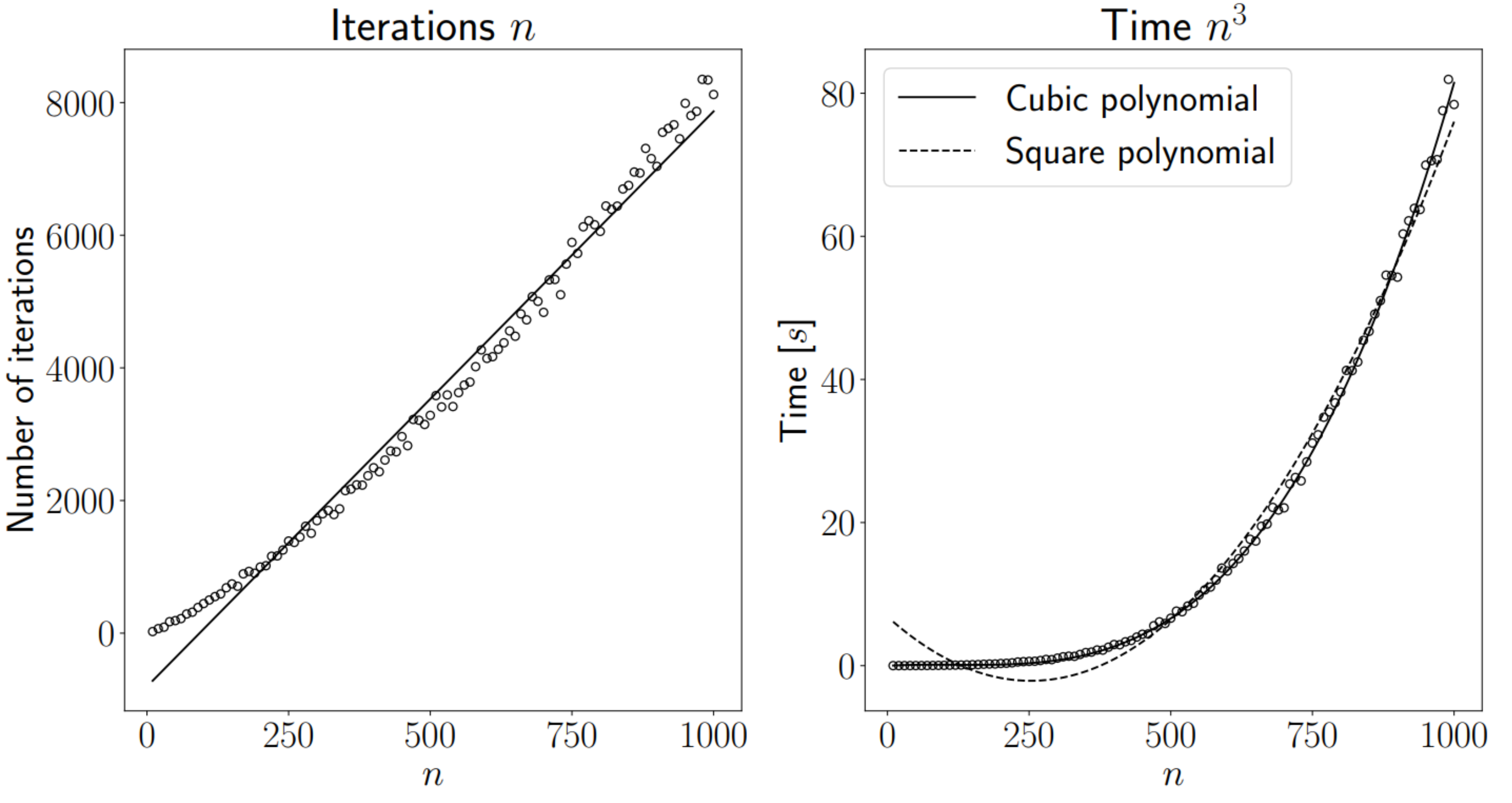
The practical performance of Simplex is very good. On average, it stops in $n$ iterations

# *Average complexity of Simplex*

**Solving random LPs to assess its practical performance**

$$\min c^T x$$
$$\text{s.t. } Ax \le b$$

$n$ variables
$3n$ constraints



# *References*

- Bertsimas/Tsitsiklis: Introduction to Linear Optimization
  - Ch 3 – The Simplex method
- Vanderbei: Linear Programming – Foundations and Extensions
  - Ch. 3 – Degeneracy
  - Ch. 4 – Efficiency of simplex method
  - Ch. 8 – implementation issues

# *Next lecture*

- Duality